



Phenotypic Screen C. Elegans Gene Pathway Identification

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Introduction

Motivation: To research C. elegans to help improve our understanding of genetic function and can contribute to advances in human disease research and drug development

Goal: Using clustering methods to identify biologically meaningful patterns in C. elegans phenotypic data and uncover potential gene pathways or functional relationships

Approach: Applied multiple clustering methods to identify structure in high-dimensional phenotypic data, evaluated model performance using internal metrics, and implemented hierarchical clustering to support biological interpretation.

Data: Consisted of proportions of 86 deformities across gene knockdowns. Data were cleaned and normalized, then dimensionality reduction (UMAP/t-SNE) was applied to mitigate noise prior to clustering. This was accomplished via R-studio

Methodology

Clustering Methods:

SNN-Leiden: graph-based clustering pipeline that identifies dense biological communities by reinforcing topological similarity and optimizing network modularity.

GMM: probabilistic clustering method that assumes each cluster has an underlying Gaussian distribution

Consensus clustering: Measures how consistent results are across multiple runs (how much certain genes are grouped together)

SNN-Louvain: Graph-based clustering method that detects communities by maximizing modularity.

Dimension Reduction Methods:

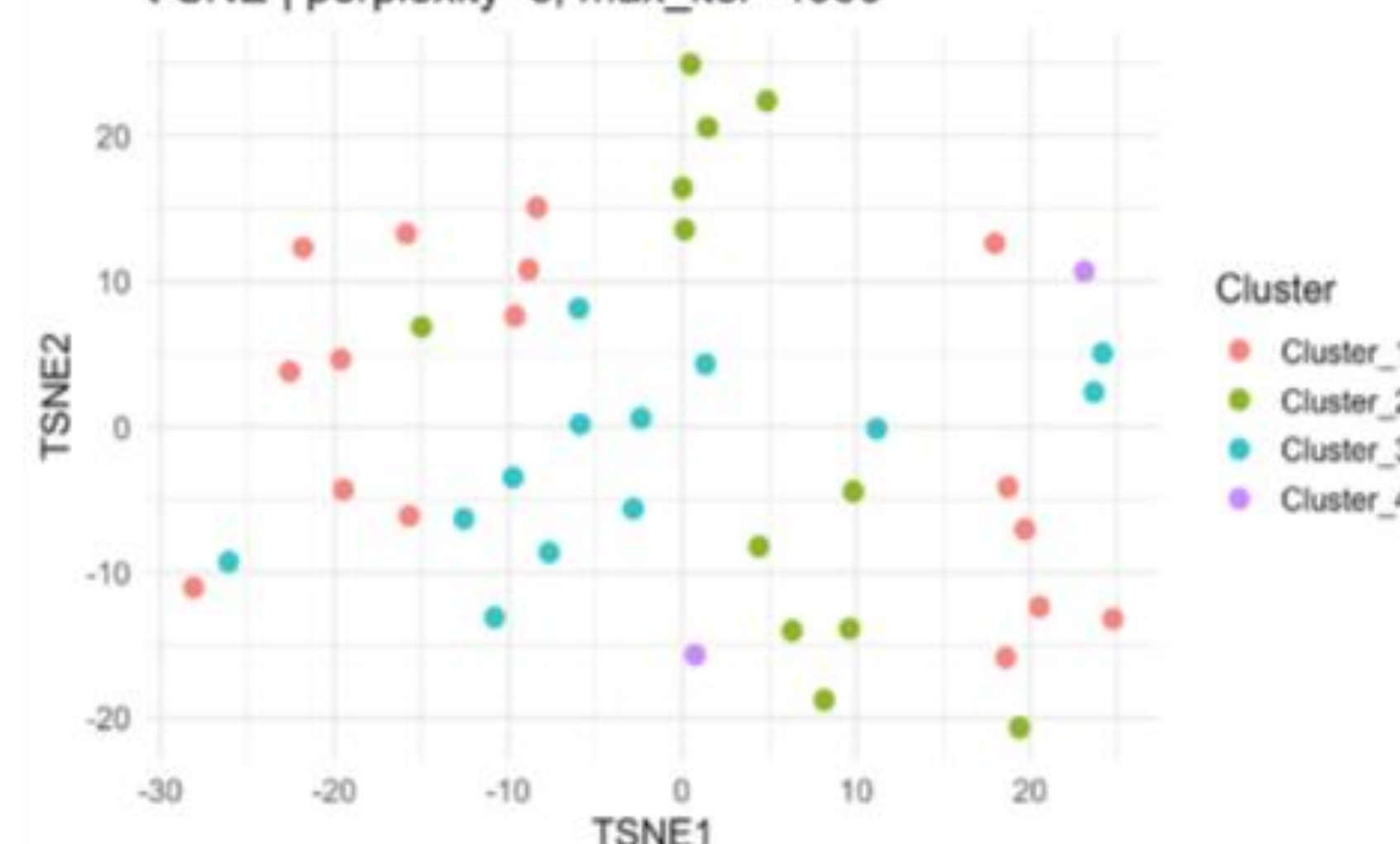
UMAP: Reduces data to low dimensions while preserving local clusters.

T-SNE: A faster dimensionality reduction method that preserves both local and global data structure.

Visualizing the Clusters

Consensus Clustering t-SNE:

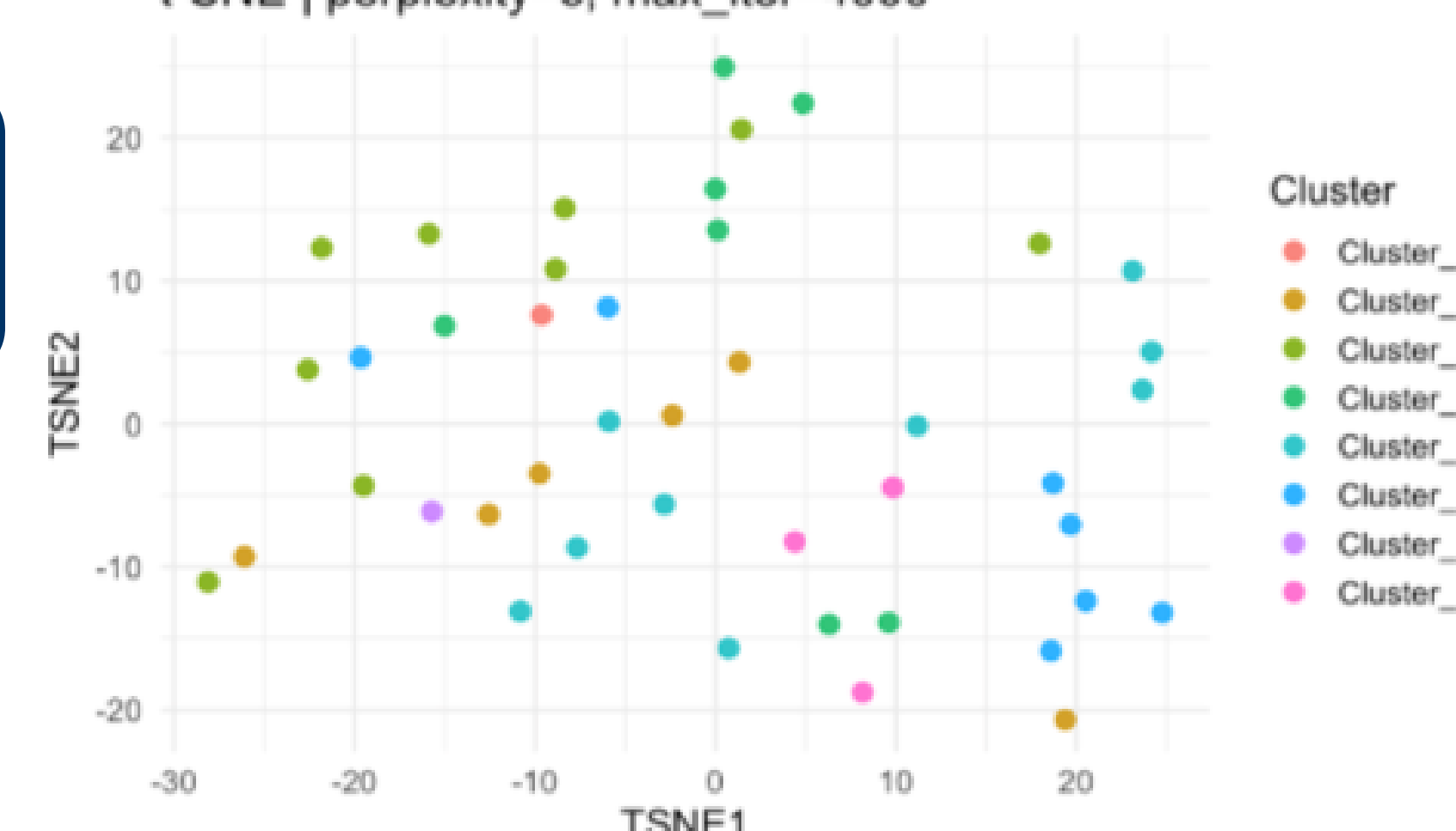
t-SNE | perplexity=5, max_iter=4000



- Clear grouping into 4 clusters (k = 4)
- Most stable assignments across runs
- Moderate separation with some overlap
- Best overall balance of structure and consistency

Leiden Clustering t-SNE:

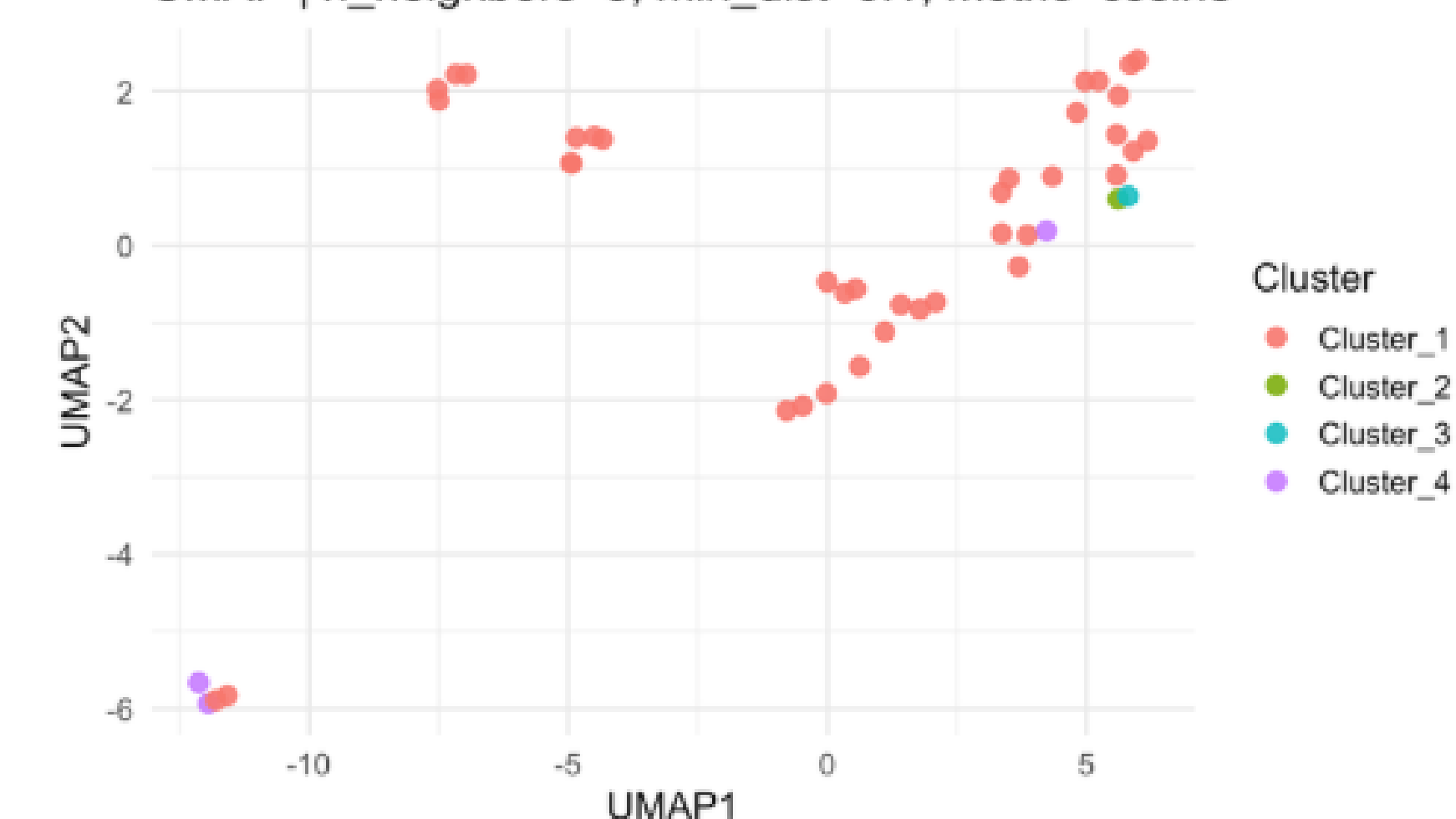
t-SNE | perplexity=5, max_iter=4000



- Produces more clusters (higher granularity)
- Clusters appear more fragmented
- Less clear separation between groups
- More sensitive to parameter choices

GMM Clustering UMAP:

UMAP | n_neighbors=3, min_dist=0.1, metric=cosine

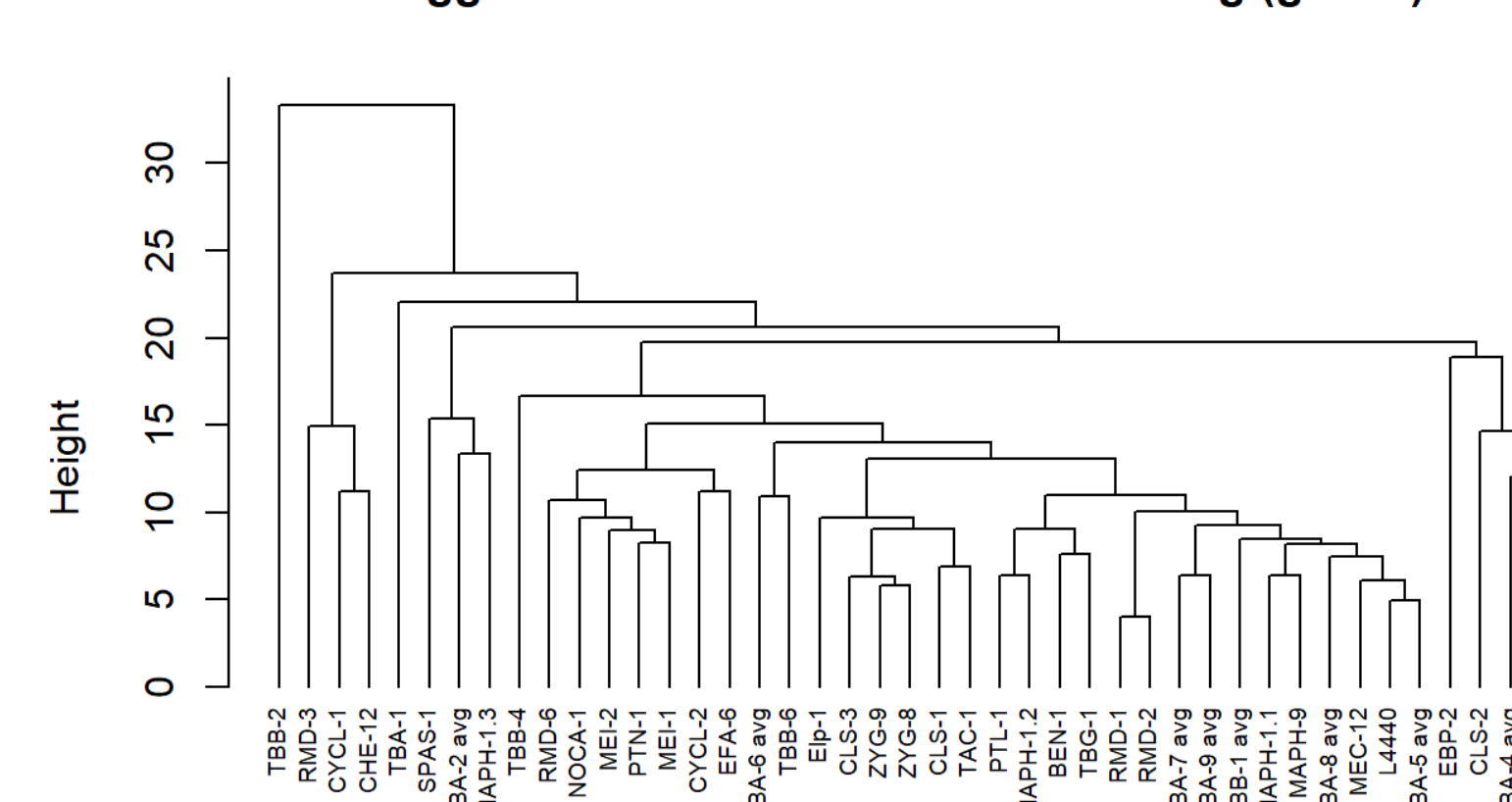


- One dominant cluster with smaller subgroups
- Some isolated outliers present
- Less balanced clustering compared to other methods

Hierarchical Clustering

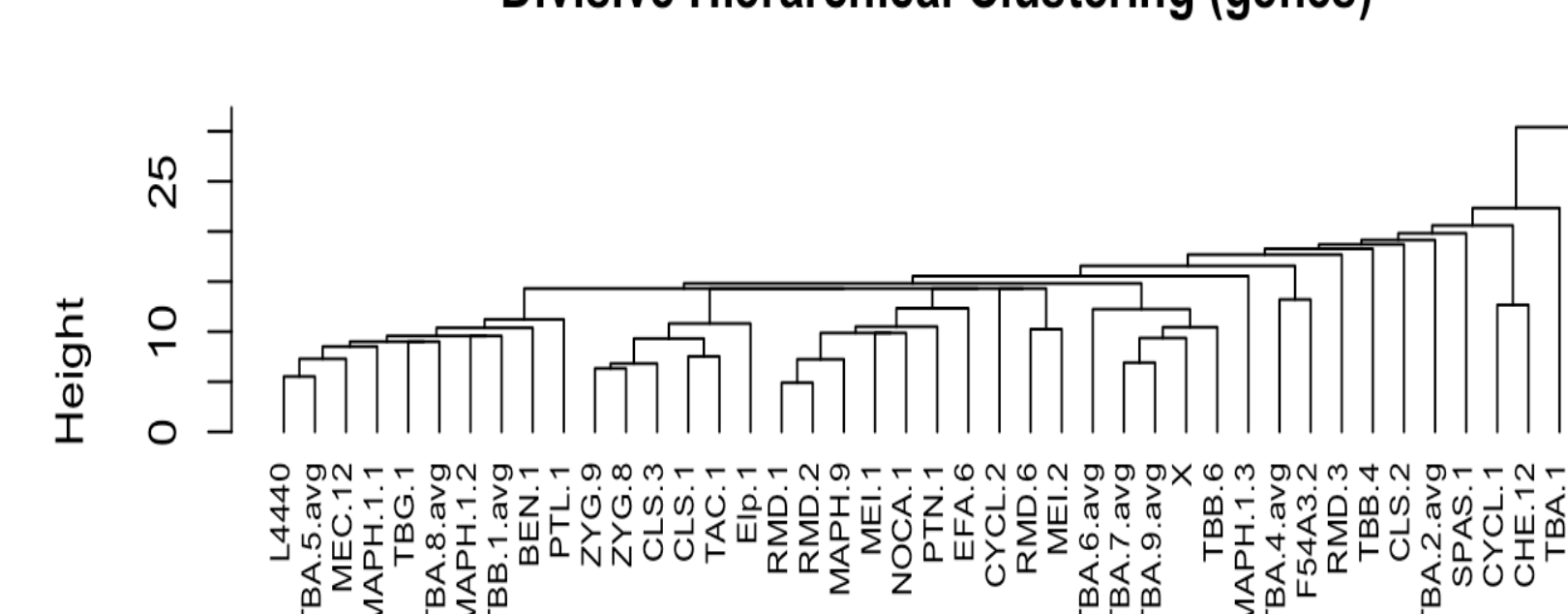
Hierarchical clustering was used to group observations based on similarity in gene expression patterns, allowing us to identify biologically meaningful structure within the data. Unlike methods that require a preset number of clusters, this approach builds a dendrogram that visualizes relationships among samples at multiple levels. Both agglomerative and divisive strategies were considered to evaluate how clusters form through merging or splitting processes. This method provides an interpretable framework for exploring complex biological datasets where statistical significance may not always reflect biological relevance.

Agglomerative Hierarchical Clustering (genes)



- Shows clustering forming through stepwise merging
- Clear grouping at higher levels of the tree
- Some clusters merge early, indicating similarity

Divisive Hierarchical Clustering (genes)



- Starts off as one group and splits into smaller clusters
- More gradual separation across samples
- Separates groups by repeatedly splitting the most dissimilar samples

Results

Clustering methods identified some underlying structure in the data, but overall cluster separation was weak, as reflected by low internal validation scores. Different methods produced varying groupings, indicating limited stability and consistency. Hierarchical clustering provided interpretable dendrograms that may help guide biological exploration despite the lack of strong statistical separation

Model	Clusters	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑
GMM	4	0.11330	1.31481	5.47540
Leiden	8	-0.25450	2.57490	1.16846
Louvain	13	-0.13942	1.89060	1.31795
Consensus	4	0.01565	2.85399	3.28628

- GMM shows highest silhouette score but weaker stability
- Leiden produces many clusters with moderate performance
- Consensus clustering is the most stable despite lower metric scores

Conclusions & Future Directions

Conclusions: Clustering methods revealed some structure in the phenotypic data, but results were not consistent or strong enough to draw definitive biological conclusions. Hierarchical clustering provides a useful framework for domain experts to explore potential gene relationships.

Future: Analyzing a larger and complete dataset, refining preprocessing and feature representation, and collaborating with biological experts to validate clusters and identify meaningful gene pathways

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